

Active Spectral Clustering via Iterative Uncertainty Reduction

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with

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15th of August 2012

Spectral Clustering

- ▶ Cluster data using only pairwise similarities.

Spectral Clustering

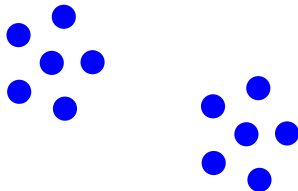
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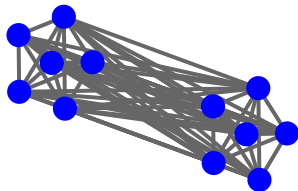
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Data

Spectral Clustering

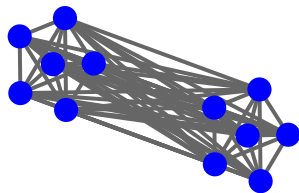
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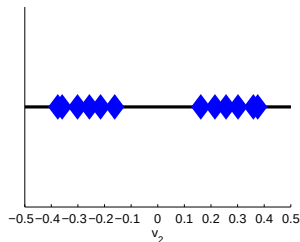
Similarities

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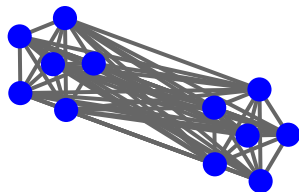
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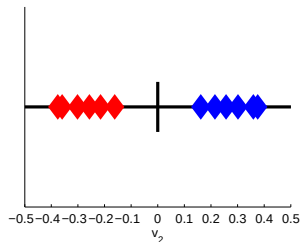
Embedding

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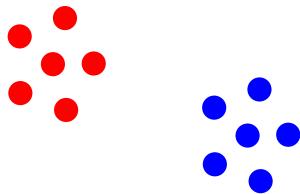
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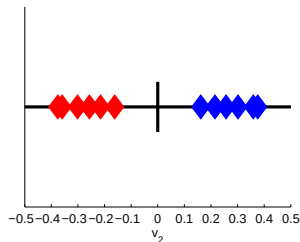
Threshold

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Clustering



Threshold

Spectral Clustering with Missing Data

Problem:

- ▶ Similarities often expensive or noisy.

Spectral Clustering with Missing Data

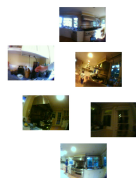
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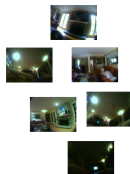
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Kitchen

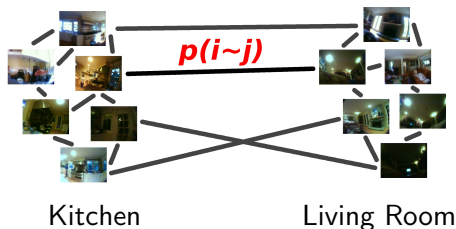


Living Room

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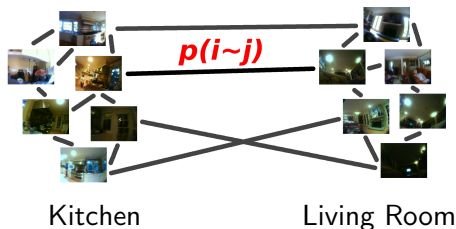
- ▶ Requires similarities reflecting co-location: e.g.

$$w_{ij} = p(i \sim j) = p(i \text{ and } j \text{ were taken in same room})$$

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- ▶ Need human annotation. Expensive/noisy. Does not scale!

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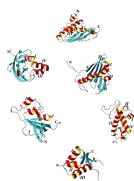
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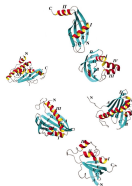
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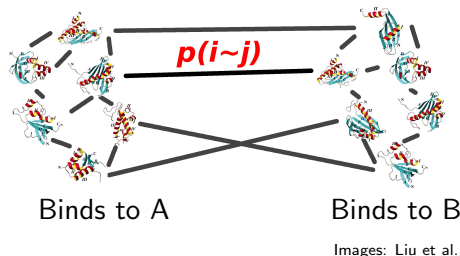
Binds to B

Images: Liu et al.

Spectral Clustering with Missing Data

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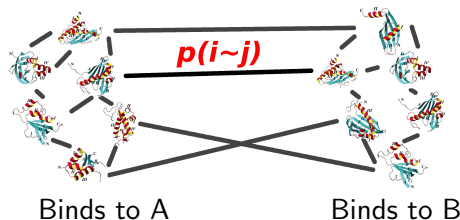
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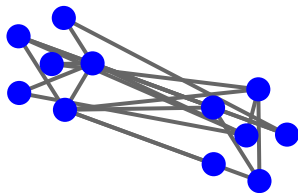
- ▶ Need experimental data. Expensive/noisy. Does not scale!

Spectral Clustering with Missing Data

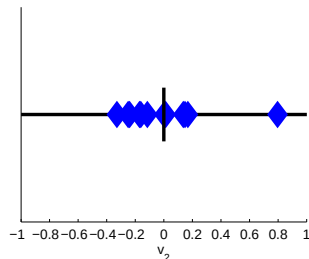
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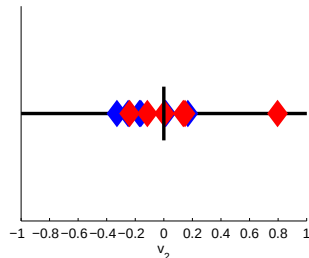
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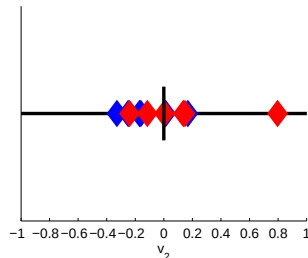
Similarities



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Similarities

Embedding

This talk: Actively measure similarities to minimize measurement cost while achieving good clustering performance.

Overview

Introduction

Spectral Embeddings

Active Clustering

Active Clustering with Noisy Similarities

Overview

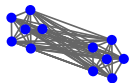
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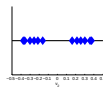
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Spectral Embeddings

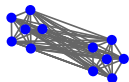


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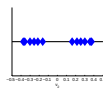


Embedding

Spectral Embeddings



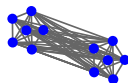
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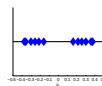
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- Given similarity matrix W , define $L = \text{diag}(W\mathbf{1}) - W$.

Spectral Embeddings



Similarities



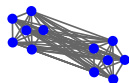
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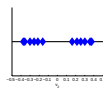
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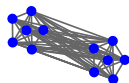
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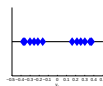
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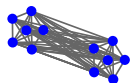
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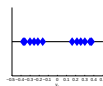
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Spectral Embeddings



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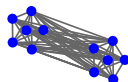
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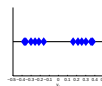
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Spectral Embeddings



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- ▶ Can use formulation to guide active learning.
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- ▶ **But data clusters, so v^* structured. Can do better!**

Spectral Embeddings

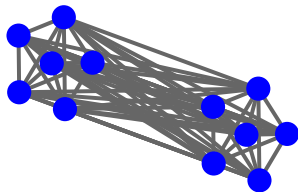
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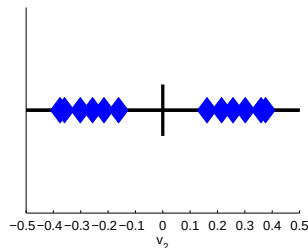
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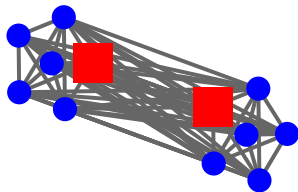
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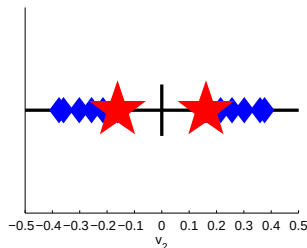
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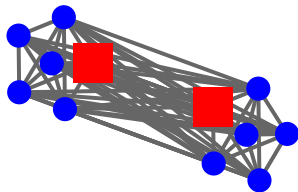
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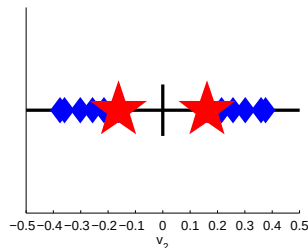
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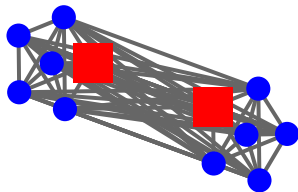


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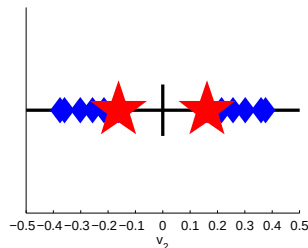
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Embedding

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Idea:

- ▶ **Actively** measure similarities to **reduce uncertainty**.

Overview

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Active Clustering

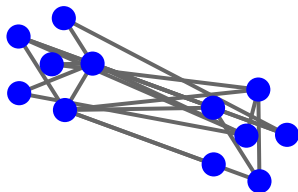
Active Clustering with Noisy Similarities

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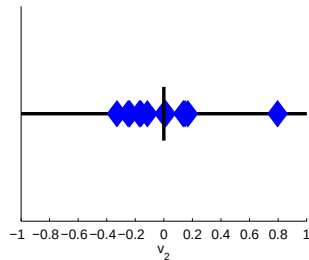
- Iteratively measure similarities which can most change the embedding **near the threshold**.

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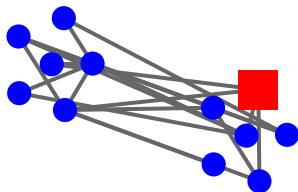
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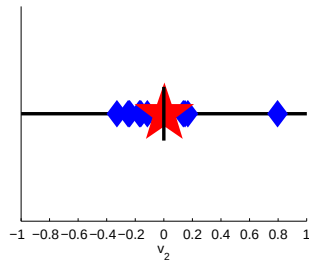
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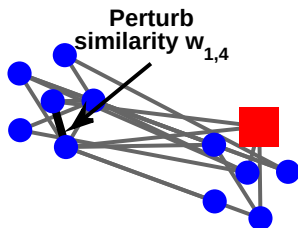
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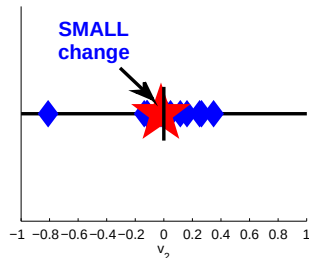
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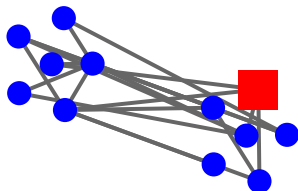
$$\left| \frac{d \star}{dw_{1,4}} \right|_0 = \text{small}$$



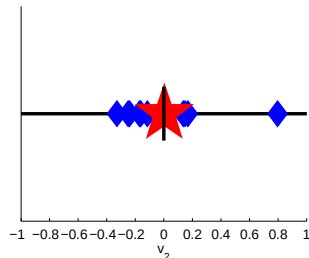
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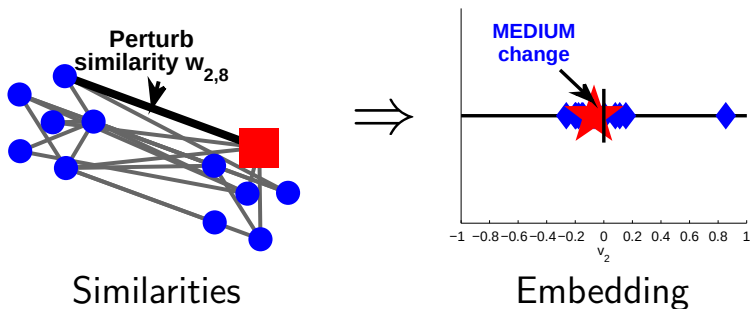


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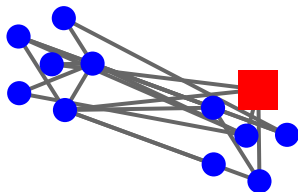
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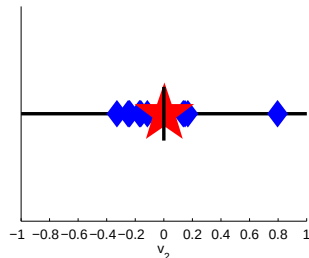
$$\left| \frac{d \star}{dw_{1,4}} \right|_0 = \text{small} \quad \left| \frac{d \star}{dw_{2,8}} \right|_0 = \text{medium}$$

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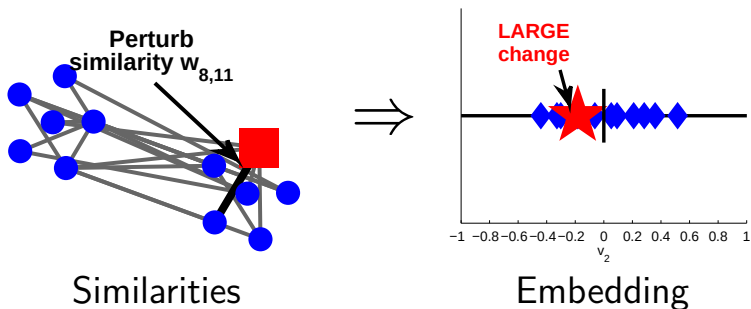


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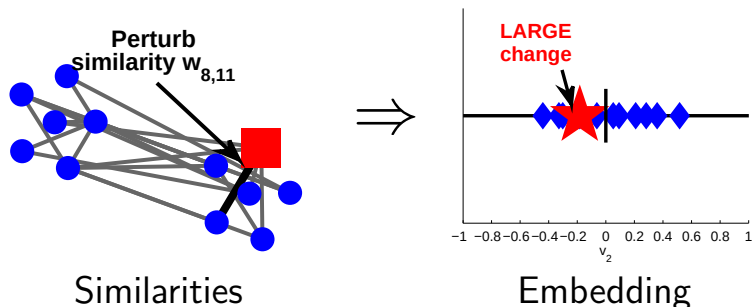
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- $\Rightarrow$ Similarity $w_{8,11}$ is **most influential**. Measure it next.

Matrix Perturbation Theory

- Recall spectral embedding:

$$\begin{aligned} v^* &= \operatorname{argmin}_v v^\top L v \\ \text{s.t. } v^\top v &= 1 \quad v^\top \mathbf{1} = 0. \end{aligned}$$

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- If $k_{\min} = \operatorname{argmin}_i |v^*(i)|$, then

$$\begin{aligned} \star &= v^*(k_{\min}) \\ \left. \frac{d \star}{dw_{ij}} \right|_0 &= \left. \frac{dv^*(k_{\min})}{dw_{ij}} \right|_0. \end{aligned}$$

Matrix Perturbation Theory

- Recall spectral embedding:

$$\begin{aligned} v^* &= \operatorname{argmin}_v v^\top L v \\ \text{s.t. } v^\top v &= 1 \quad v^\top \mathbf{1} = 0. \end{aligned}$$

- Embedding v^* is eigenvector of L .
- If $k_{\min} = \operatorname{argmin}_i |v^*(i)|$, then

$$\begin{aligned} \star &= v^*(k_{\min}) \\ \left. \frac{d\star}{dw_{ij}} \right|_0 &= \left. \frac{dv^*(k_{\min})}{dw_{ij}} \right|_0. \end{aligned}$$

- **Matrix Perturbation Theory:**

$$\left. \frac{dv^*(k_{\min})}{dw_{ij}} \right|_0 = \sum_{p>2}^n \frac{v_2^\top [\partial L / \partial w_{ij}] v_p}{\lambda_2 - \lambda_p} v_p(k_{\min}).$$

Algorithm Sketch

Impute unobserved similarities in W with 0.

Iterate:

1. Compute embedding v^* .
2. $(i^*, j^*) = \operatorname{argmax}_{(i,j) \in \text{Unobserved}} \left| \frac{dv^*(k_{\min})}{dw_{ij}} \right|_0$.
3. Measure w_{i^*, j^*} and add to W .

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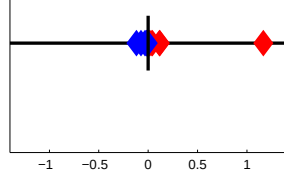
00 measurements out of 66



Similarities



00 measurements out of 66



Embedding

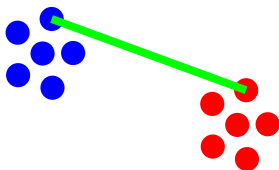
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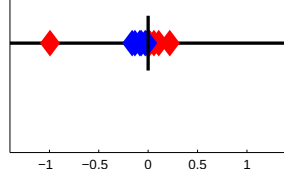
01 measurements out of 66



Similarities



01 measurements out of 66



Embedding

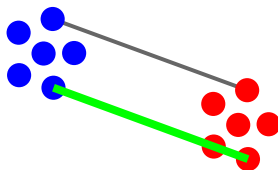
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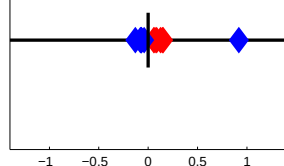
02 measurements out of 66



Similarities



02 measurements out of 66



Embedding

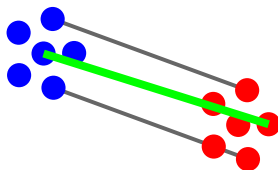
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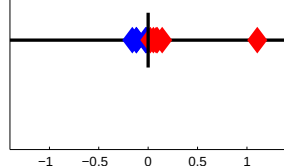
03 measurements out of 66



Similarities



03 measurements out of 66



Embedding

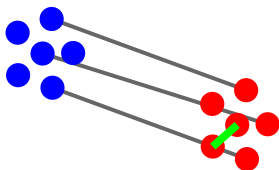
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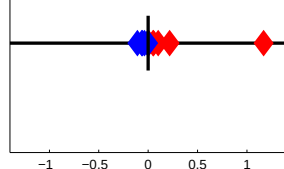
04 measurements out of 66



Similarities



04 measurements out of 66



Embedding

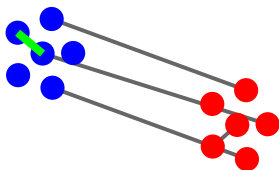
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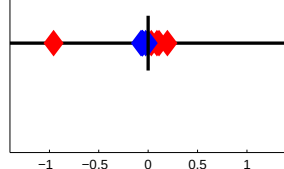
05 measurements out of 66



Similarities



05 measurements out of 66



Embedding

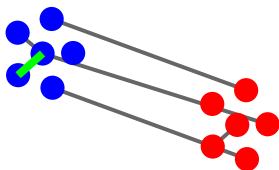
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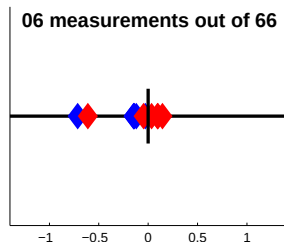
06 measurements out of 66



Similarities



06 measurements out of 66



Embedding

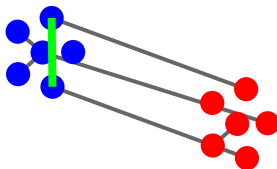
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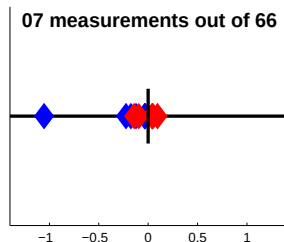
07 measurements out of 66



Similarities



07 measurements out of 66



Embedding

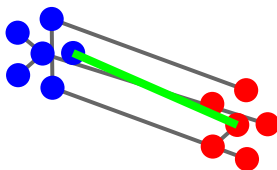
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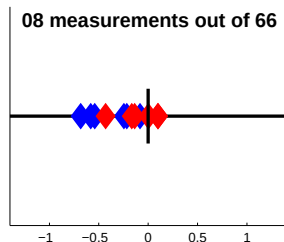
08 measurements out of 66



Similarities



08 measurements out of 66



Embedding

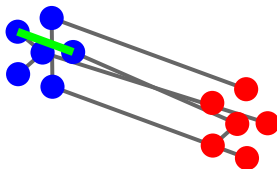
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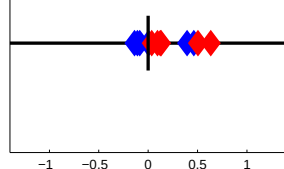
09 measurements out of 66



Similarities



09 measurements out of 66



Embedding

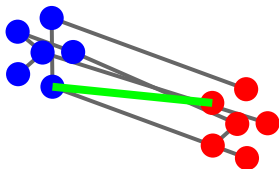
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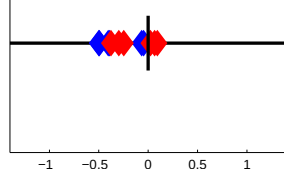
10 measurements out of 66



Similarities



10 measurements out of 66



Embedding

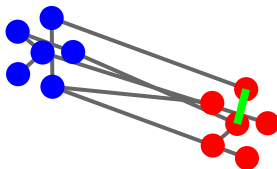
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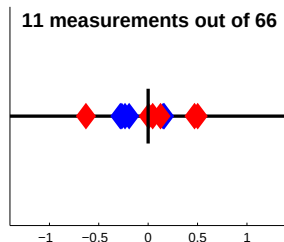
11 measurements out of 66



Similarities



11 measurements out of 66



Embedding

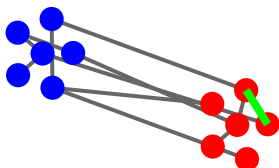
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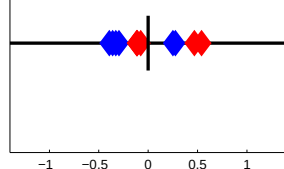
12 measurements out of 66



Similarities



12 measurements out of 66



Embedding

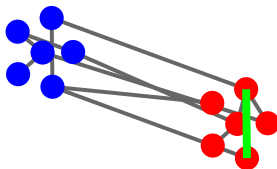
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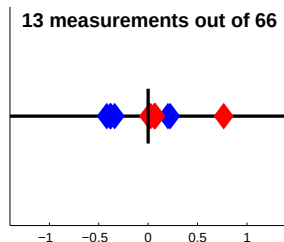
13 measurements out of 66



Similarities



13 measurements out of 66



Embedding

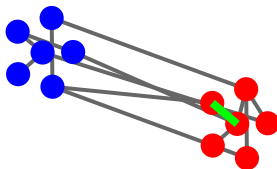
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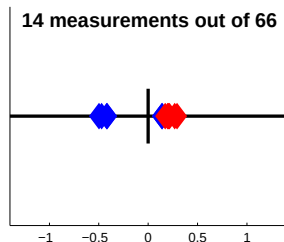
14 measurements out of 66



Similarities



14 measurements out of 66



Embedding

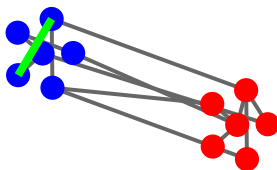
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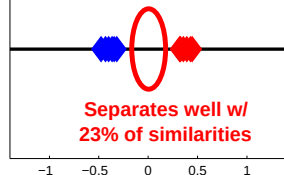
15 measurements out of 66



Similarities



15 measurements out of 66



Embedding

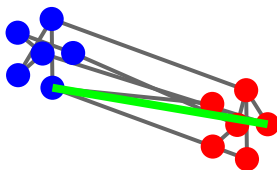
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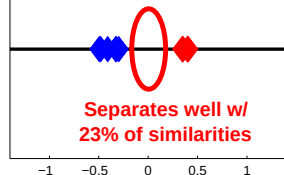
16 measurements out of 66



Similarities



16 measurements out of 66



Embedding

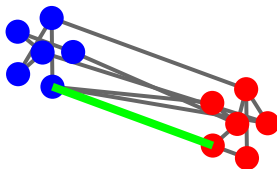
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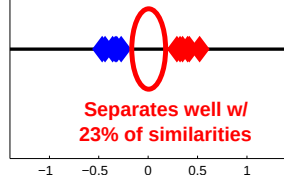
17 measurements out of 66



Similarities



17 measurements out of 66



Embedding

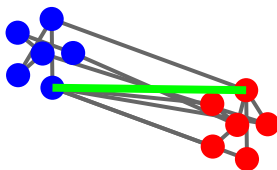
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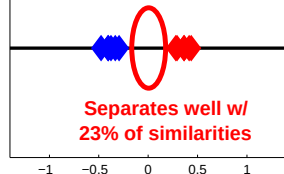
18 measurements out of 66



Similarities



18 measurements out of 66



Separates well w/
23% of similarities

Embedding

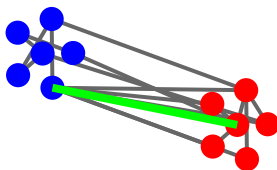
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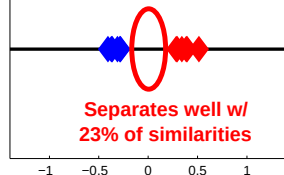
19 measurements out of 66



Similarities



19 measurements out of 66



Embedding

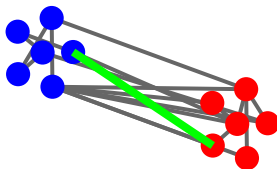
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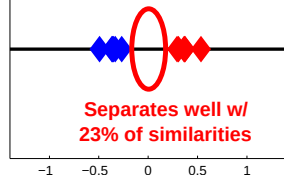
20 measurements out of 66



Similarities



20 measurements out of 66



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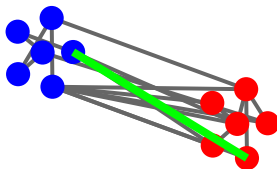
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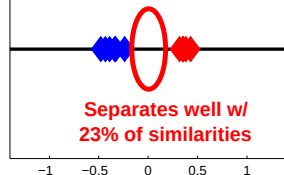
21 measurements out of 66



Similarities



21 measurements out of 66



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Embedding

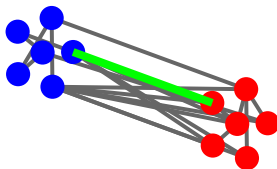
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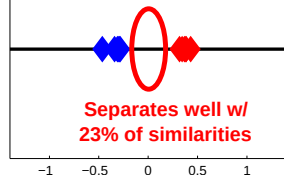
22 measurements out of 66



Similarities



22 measurements out of 66



Embedding

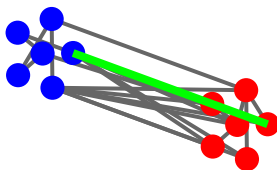
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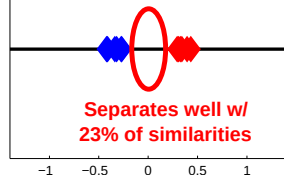
23 measurements out of 66



Similarities



23 measurements out of 66



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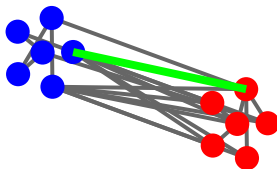
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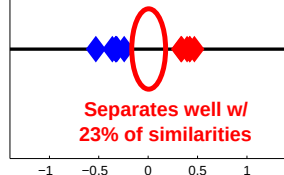
24 measurements out of 66



Similarities



24 measurements out of 66



Embedding

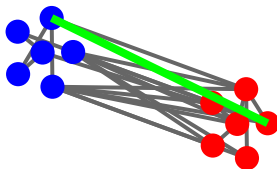
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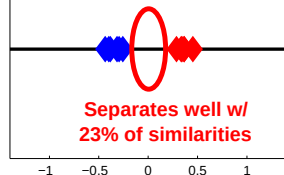
25 measurements out of 66



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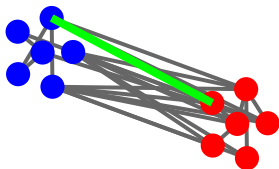
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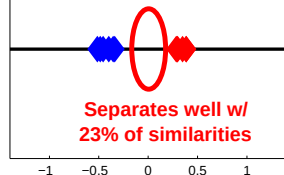
26 measurements out of 66



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26 measurements out of 66



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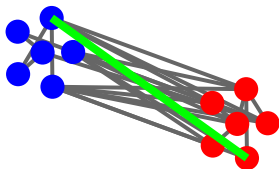
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2. $(i^*, j^*) = \operatorname{argmax}_{(i,j) \in \text{Unobserved}} \left| \frac{dv^*(k_{\min})}{dw_{ij}} \right|_0$.
3. Measure w_{i^*, j^*} and add to W .

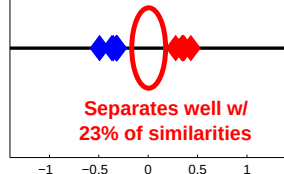
27 measurements out of 66



Similarities



27 measurements out of 66



Separates well w/
23% of similarities

Embedding

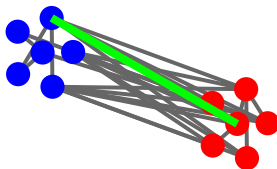
Algorithm Sketch

Impute unobserved similarities in W with 0.

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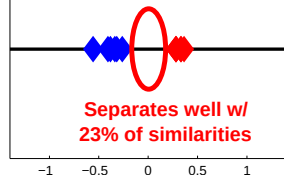
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Embedding

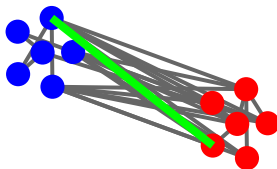
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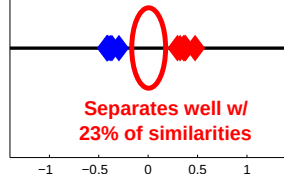
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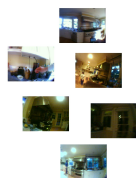
29 measurements out of 66



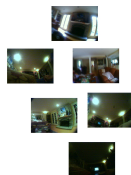
Separates well w/
23% of similarities

Embedding

Application: Clustering Photos by Location



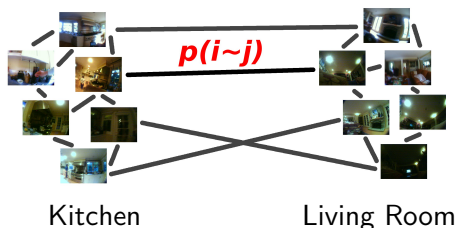
Kitchen



Living Room

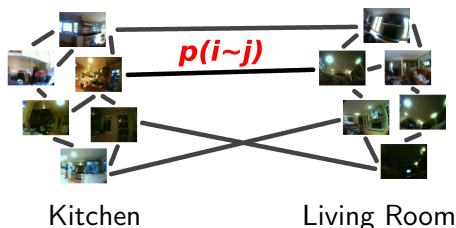
- Cluster 100 photos into kitchen/living room.

Application: Clustering Photos by Location



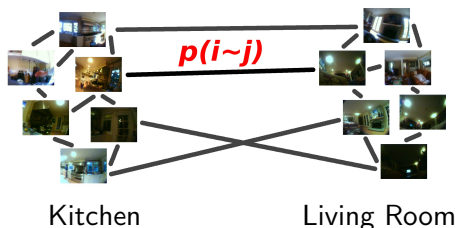
- ▶ Cluster 100 photos into kitchen/living room.
- ▶ Similarities $w_{ij} = p(i \sim j)$ are HITs on Mechanical Turk:
How likely is it the photos were taken in the same room?

Application: Clustering Photos by Location



- ▶ Cluster 100 photos into kitchen/living room.
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How likely is it the photos were taken in the same room?
- ▶ Similarities are median of **three** noisy HIT responses.

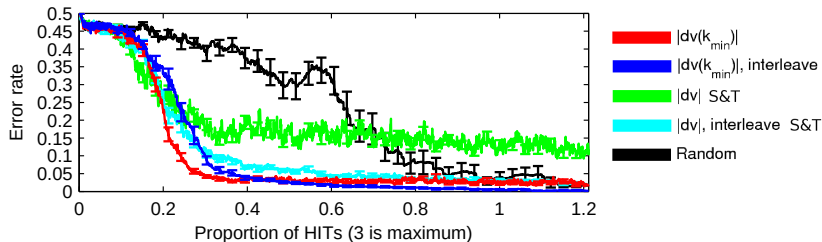
Application: Clustering Photos by Location



- ▶ Cluster 100 photos into kitchen/living room.
- ▶ Similarities $w_{ij} = p(i \sim j)$ are HITs on Mechanical Turk:
How likely is it the photos were taken in the same room?
- ▶ Similarities are median of **three** noisy HIT responses.
- ▶ Can cluster with all similarities. Expensive (US\$222)!

Application: Clustering Photos by Location

- Proportion p means $p \binom{n}{2}$ of $3 \binom{n}{2}$ HIT responses used.



- S&T need US\$32 to reach error rate 0.05. We need US\$17!

Overview

Introduction

Spectral Embeddings

Active Clustering

Active Clustering with Noisy Similarities

Noisy Similarities

- ▶ Can use median of repeat measurements to reduce noise.
- ▶ Influence of noise on embedding captured by $\left| \frac{dv^*(k_{\min})}{dw_{ij}} \right|_0$.
⇒ Only need to know influential similarities accurately.

Noisy Similarities

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At each measurement have choice. Can reduce

- ▶ noise by re-measuring similarities.
- ▶ cluster uncertainty by measuring most influential similarity.

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 $\Rightarrow$ Only need to know influential similarities accurately.

At each measurement have choice. Can reduce

- ▶ noise by re-measuring similarities.
- ▶ cluster uncertainty by measuring most influential similarity.

Idea:

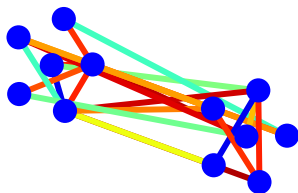
- ▶ Measure the most uncertain and influential similarity.
- ▶ Augment algorithm to estimate:
 - running medians $\bar{w}_{ij}$ of similarities
 - standard deviations σ_{ij} of medians (frequentist or Bayesian)
- ▶ Measure similarity w_{ij} where $\left| \frac{dv^*(k_{\min})}{dw_{ij}} \right|_{\bar{w}_{ij}} \cdot \sigma_{ij}$ is largest

Active Clustering with Noisy Similarities

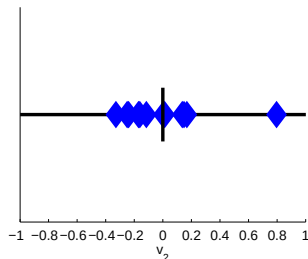
- ▶ Measure the most **uncertain** and **influential** similarity.

Active Clustering with Noisy Similarities

- Measure the most **uncertain** and **influential** similarity.



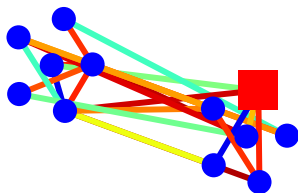
Similarities



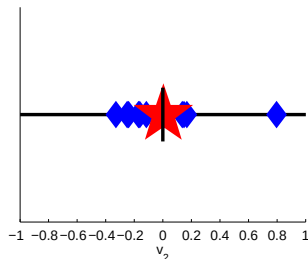
Embedding

Active Clustering with Noisy Similarities

- Measure the most **uncertain** and **influential** similarity.



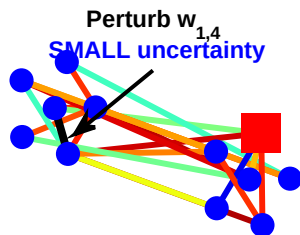
Similarities



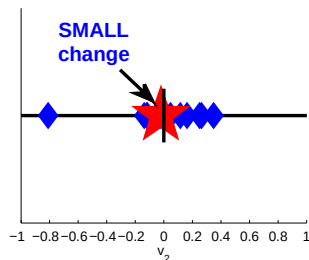
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Active Clustering with Noisy Similarities

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Similarities

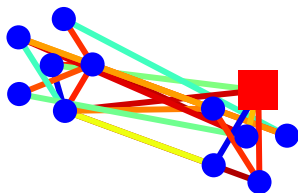


Embedding

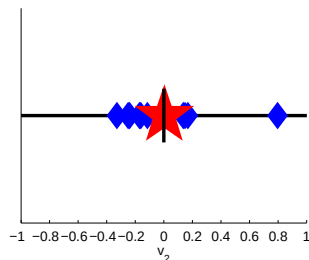
$$\left| \frac{d \star}{dw_{1,4}} \right|_{\bar{w}_{1,4}} \sigma_{1,4} = \text{small}$$

Active Clustering with Noisy Similarities

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Similarities

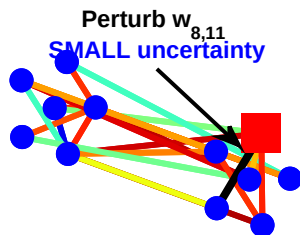


Embedding

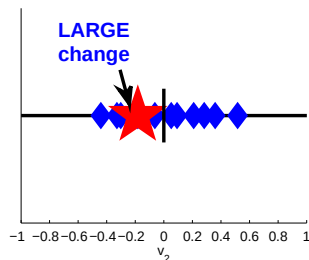
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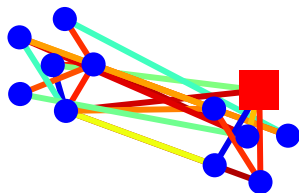


Embedding

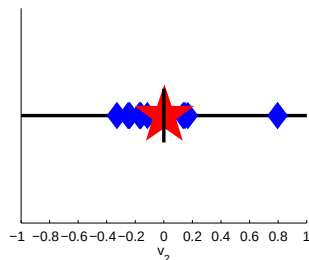
$$\left| \frac{d \star}{dw_{1,4}} \right|_{\bar{w}_{1,4}} \sigma_{1,4} = \text{small} \quad \left| \frac{d \star}{dw_{8,11}} \right|_{\bar{w}_{8,11}} \sigma_{8,11} = \text{med.}$$

Active Clustering with Noisy Similarities

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Similarities

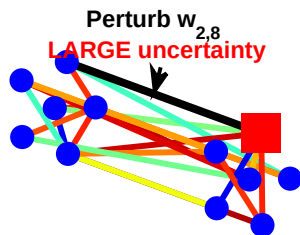


Embedding

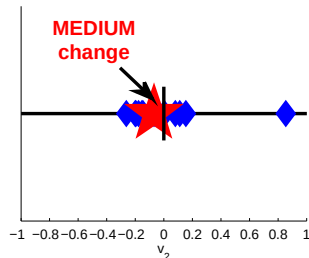
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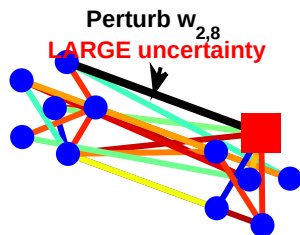


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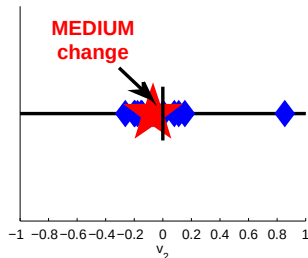
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Active Clustering with Noisy Similarities

- Measure the most **uncertain** and **influential** similarity.



Similarities



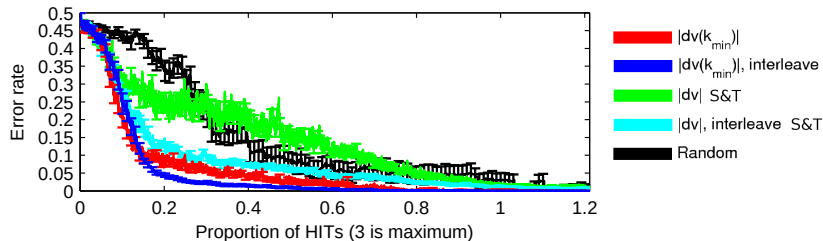
Embedding

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- $\Rightarrow w_{2,8}$ **most uncertain and influential.** Measure it next.

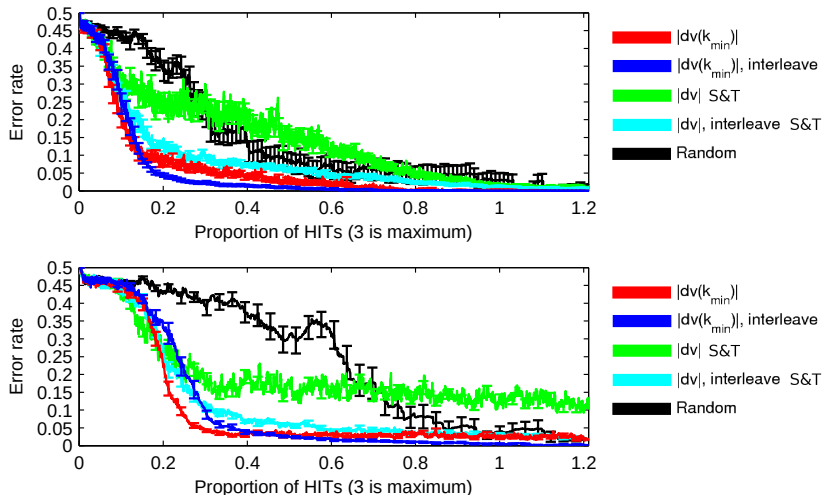
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Questions?

